

Instruction Manual

Operating the Transit Calculator

Transit Calculator

Version 1.0 · 14 August 2026

Who this is for

The operator planning a passage. Assumes you have the tool running — see the Quick Start if not. Covers every control, and what each result means.

1 The screen

Three regions: the chart fills the window, a panel at the left builds the line, a panel at the right computes and exports it. A status bar along the bottom carries the cursor position, the zoom, and what is cached.

Region	What lives there
Chart	NOAA ENC tiles, the transit line, waypoints, numbered legs, direction chevrons and per-leg current arrows once computed. Scroll to zoom; drag to pan.
Left panel	Transit line (draw / undo / clear), Load a line, REVERSE, and Localize for offline.
Right panel	Passage settings, CALCULATE, the results, and Export.
Status bar	Cursor position, zoom level, tiles held, current cycle, and a running summary of the drawn line.

2 Building a line

2.1 Drawing

DRAW mode is on by default. Click the chart to drop points in order. PAN mode drags the chart without adding points.

Action	How
Add a point	Click the chart (DRAW mode)
Move a point	Drag it
Select a point	Click it
Delete a point	Click it, then press Delete
Remove the last	Undo
Discard the line	Clear
Run it the other way	REVERSE, at the foot of the panel

Every control in the console carries a tooltip — the buttons, the fields, and the column headings of the leg table. So does every abbreviation it puts on screen: DBOFS and the other model names, the chips above the leg table, IDW, NDBC, WW3, and the cycle tag in the status bar are all written out in full on hover. Hover rather than coming back here.

2.2 Loading

Pick a file from the list and press Load, or Upload to browse. Supported:

Format	Notes
Shapefile (.shp)	Needs its .prj to be placed correctly. UTM and geographic both read. A .prj-less file whose coordinates are not lat/lon is REFUSED rather than guessed at — a guessed zone puts the line in the wrong ocean.
Zipped shapefile	Drop the whole set in as one .zip.
GeoJSON	The longest LineString is used.
GPX	Route points or track points.
CSV	Needs latitude/longitude columns (lat/lon, y/x also accepted).

After loading, the panel reports how many points came in, the CRS it found, and anything it skipped. A null shape — what QGIS leaves when a feature is deleted without repacking — is counted and skipped, not treated as corruption.

2.3 Direction

Which way the line runs changes the answer, so it is shown on the chart rather than left to be inferred: the ends are labelled START and END, and chevrons along each leg point the way of travel. REVERSE sits at the foot of the panel, below the loading controls and immediately above the offline downloads — it is the last decision about the geometry before you fetch data for it.

REVERSE reverses the geometry, not a label

The leg table, the ETAs and the exported shapefile all describe the run that was computed. There is no second convention to keep straight. The previous result is cleared rather than left on screen describing the other direction.

3 Localizing for offline

Both downloads are cut to the extent of the line you have drawn, so they stay small. Draw or load the line first.

Button	What it fetches	Size
Charts	NOAA ENC raster tiles over the line, zoom 8 to 12.	Tens of MB, depending on the area
Currents	Works out which forecast models actually reach the line and fetches each one, scoped to the line.	About 6 to 13 MB per model when scoped

Before downloading, the Currents button reports which models the line needs and what share of it each covers, so you can see the cost before paying it. A tick means that model is already cached.

4 Passage settings

4.1 Speed, and the three things it can mean

This is the setting most likely to be misread, so it is worth being precise about.

Mode	You give it	The weather and current then change
through water	A water speed to hold — how fast the hull moves through the water, which is what the engine actually controls.	Your speed over ground, and therefore the ETA. Weather is paid for in revs, so it costs fuel and not time.
over ground	A speed made good, i.e. you have an arrival time to hit.	The water speed needed, the RPM, and therefore the fuel.
fixed revs	The throttle setting itself. The speed is then an OUTPUT.	The speed through the water, and so the ETA. Litres per hour is fixed by the revs, so weather costs TIME here rather than fuel.

Fixed revs is not "through water" by another name

They agree only if the vessel's own controller agrees with this model about which revs give which speed. That is worth checking rather than assuming: an ASV commanded to hold a speed converts it to revs using ITS internal model, and if that model is optimistic the boat sits at fewer revs and goes slower than the number on the console says. Running the same plan in both modes — one at the commanded speed, one at the observed revs — shows the size of any disagreement directly.

4.2 Departure

The tide is not the same three hours later. Departure is a real input, not a label: the calculation looks up the current at the position the vessel has reached and the clock time it got there. It defaults to the start of the cached forecast, not to "now", because a cycle fetched yesterday cannot answer for this afternoon and defaulting to now would quietly push every leg onto the projection path.

4.3 Fuel aboard

Litres in the tank at departure. Leave it blank for a full tank, which is what the tool assumed before this field existed. It changes nothing about the passage — the same line burns the same fuel — only what that burn is measured against.

The reserve floor belongs to the tank, not to the load

The floor is 25% of the 250 L tank, so 62.5 L, and it does not shrink when you sail part-full. Enter 140 L aboard and 77.5 L is yours to burn — the whole 110 L shortfall comes out of usable fuel, not a quarter of it. Taking the reserve as a fraction of what is loaded would report 105.0 L instead, and err in the direction that runs a boat dry.

A figure larger than the tank holds is refused rather than trimmed to full: it is a mistyped gauge reading, and quietly trimming it would hand back a margin you never had. A load already below the reserve floor is accepted and flagged — you may be planning what to load — and reported as a distinct condition from a passage that merely eats the margin.

4.4 Currents

Leave the tick on and the selector at "all models (chained)". Selecting a single cycle forces one model, which re-opens whatever domain gap the chaining exists to close.

4.5 Weather

Setting	What happens
IDW over the line	Buoy observations and gridded forecasts are gathered along the whole line and interpolated to every step. This is the default and the most honest option.
enter below	You supply one wind and sea state for the whole passage. Useful for a what-if, or when you have a forecast the tool cannot reach.
ignore (benign)	No weather premium at all. Use only to isolate the effect of something else — it will understate fuel in any real sea.

5 Reading the result

5.1 Headline

Distance, time, litres, and average speed over ground. Distance is geometry and will not move; the other three depend on every setting above.

5.2 The leg table

Column	Meaning
NM	Geodesic length of the leg
Brg	True bearing at the start of the leg
SOG	Speed over ground made good along the track, averaged over the leg
Set / Dft	Mean current direction (toward) and speed, vector-averaged
RPM	Required revs, after the sea-state and heading premiums
Time / ETA	Leg duration and arrival time, UTC
L	Fuel for the leg
Cov	Coverage: percentage of the leg that had a real forecast current

Hover a row for speed through water, the along-track component of the current, the premium applied, and the wind and sea state used.

5.3 Finding a leg on the chart

Every leg carries its number on the chart, boxed beside the segment it names, and hovering a row in the table lights that segment white. So a row saying leg 4 cannot hold track can be found without counting segments. Where the line is too crowded for a badge to sit beside its leg — the first legs out of a harbour, at low zoom — the badge is pushed clear and a leader points back to the leg it belongs to, rather than being dropped.

A leg is a segment, a waypoint is a point

Leg 4 runs from WP4 to WP5. Six waypoints make five legs, so the two numberings do not line up past the first, and reading a leg number as a waypoint puts you one segment out.

5.4 Endurance

Fuel against what is in the tank and the reserve floor. On a full 250 L tank with a 25% reserve, 188 L is usable; the margin shown is what is left after the passage. A negative margin means the passage breaks the reserve policy, not that the vessel runs dry.

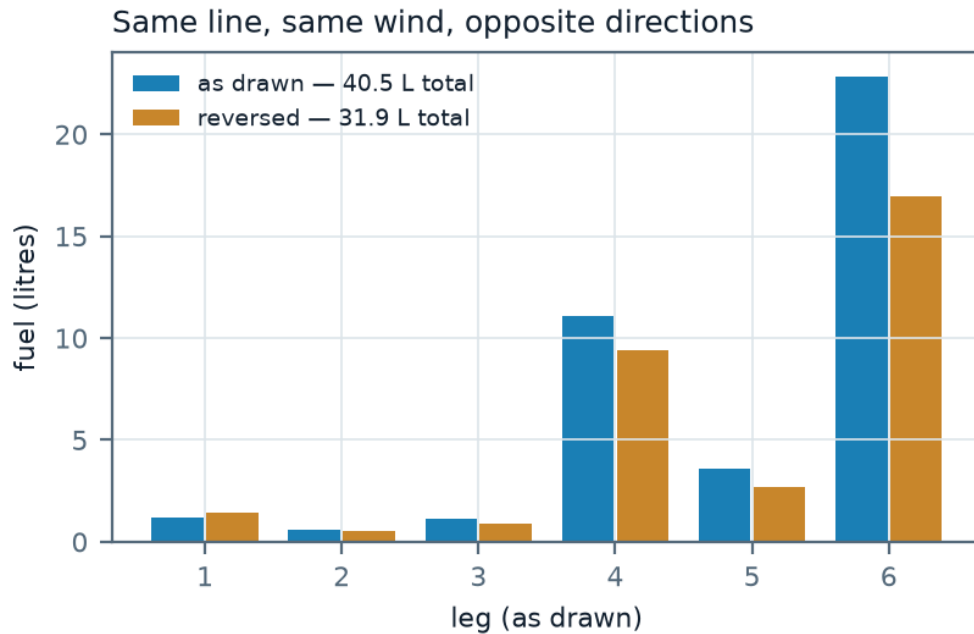
State a fuel aboard (4.3) and the line names it: the usable figure is then that load less the reserve floor, and the floor is quoted in litres rather than as a percentage — 140 L aboard of a 250 L tank leaves 78 L usable, and a 40 L passage a margin of 37.5 L. A percentage there would read as a percentage of the load, which is exactly the arithmetic the tool refuses to do.

5.5 Weather sources

Expand this to see every station and grid node that contributed, its distance, its age, and which fields it supplied. It is worth looking at: buoys report partially, and this is where you see that the wind came from one station and the waves from another.

6 Comparing directions

"Compare the other direction" computes the reversed line at the same departure and reports the difference, keeping both results. On the supplied line the two directions differ by about a fifth of the fuel, because the long leg runs into the wind one way and away from it the other, and the burn law is convex.



The same line and the same wind, run both ways. Per-leg fuel, computed by the planner at build time.

7 Emergency — predicting a drift

If the vessel loses power, the bottom of the left panel answers a different question from the rest of this tool: not where it is going, but where it is being taken. Set the last known position — press "Set last known position" and click the chart, or type the coordinates — give the time it was lost and how many hours ahead to look, and press PREDICT DRIFT.

What you get	What it means
The datum	Where the middle of every assumption puts the hull. It is not a position report and should never be passed on as one.
The radius	A circle around the datum holding every member of the ensemble. This is the spread of the stated assumptions — search area, not an error bar and not a probability.
The cloud	Every ensemble member's path, drawn faintly. Where it is wide, the answer is weak, and that is worth seeing rather than being told.
The horizon	The hour past which the forecast could not answer, and why. The track is dashed beyond it.

7.1 Where the leeway comes from

The wind term uses the US Coast Guard leeway model — the same formulation operational SAR services run — rather than a rule of thumb. Each object class carries a downwind and a crosswind regression against wind speed, and each of those carries the standard error of the field experiment that produced it:

$$DWL = (\text{slope} + \text{eps}/20) \times W10 + \text{offset} + \text{eps}/2 \quad [\text{cm/s}]$$

W10 is the wind at 10 m in m/s; crosswind is the same form with its own coefficients, taken left or right of downwind.

Three consequences worth knowing. Some classes have a non-zero OFFSET, so they drift even in no wind. The divergence angle is not a setting — it falls out of the ratio of the two regressions, and differs by object. And the spread of the search circle is the MEASURED error of those regressions, not a guess: it is where most of the radius comes from.

Class	Downwind	Crosswind	Measured spread
Person in water, unknown state	0.96%	+0.54 / -0.54%	12.0 cm/s
Self-locating datum marker buoy — no windage	0.00%	+0.00 / 0.00%	0.0 cm/s
Navy sub-escape (SEIE) 1-man raft, with drogue	1.70%	+0.00 / 0.00%	4.2 cm/s
Navy sub-escape (SEIE) 1-man raft, no drogue	3.30%	+0.50 / 0.10%	4.2 cm/s
Sea kayak with person on aft deck	1.16%	+0.41 / -0.41%	4.1 cm/s
Skiff, swamped and capsized	1.65%	+0.39 / -0.39%	3.1 cm/s
Surfboard with person	1.93%	+0.51 / -0.51%	8.3 cm/s
Skiff, modified-V cathedral-hull runabout	3.15%	+1.29 / -1.29%	2.2 cm/s

No class exists for an unmanned surface vessel

The published taxonomy is people, rafts and boats — 85 categories, none of them a USV. The default here is "Navy sub-escape (SEIE) 1-man raft, with drogue", which puts the hull at 0.26 kt downwind in a 20 kt wind. It is not a guess from hull shape: this class has been found to closely match the vessel in past operations. That is OPERATIONAL EXPERIENCE, not a measurement of this hull — a drift trial with a GPS log in a known wind would replace it, and remains the single most valuable thing that could be done for this feature. Any of the other classes can be selected if a particular casualty warrants it.

Two features of that class are worth knowing before you read a radius from it. Its crosswind does not scale with wind speed at all — it is a constant, and an asymmetric one, larger to the right of downwind than to the left. And its downwind offset is **NEGATIVE**, so in light airs it is measured drifting slightly behind the water rather than with it.

Advection — the hull going with the water — comes from the same NOAA forecast the passage uses. Leeway is defined relative to the current at about 0.5 m depth, which is how the field experiments measured it, so adding the two is the correct convention rather than double counting the wind.

7.2 Swell, and why only some of it counts

The optional swell boxes add a Stokes drift term — the transport waves carry over and above the current. Leave them blank and there is no wave term at all.

Only the swell that is NOT running with the wind

The leeway coefficients were measured in the field with seas running, so the wave forcing that goes with the local wind is already inside them, and adding a whole Stokes vector on top would count it twice. The component along the wind is projected out and only the remainder is used: a swell running with the breeze changes nothing, while an old swell across or against it counts in full. That is the case this term exists for — a calm morning with a swell left over from a distant blow.

The magnitude is the deep-water narrow-band result: a 2 m swell at 8 s gives 0.048 kt. The **PERIOD** matters more than the height, because the transport goes as the cube of frequency — a short steep sea carries several times what a long swell of the same height does. Where height and period disagree badly enough to imply a breaking wave the term is capped and the result says so, because Stokes goes as the square of steepness and bad data would otherwise invent a knot of drift.

Two limits worth knowing. Deep water is assumed, so in depths shallower than about half a wavelength this understates the transport, and the tool has no bathymetry with which to know. And the term is not varied across the ensemble: it sits an order of magnitude below the leeway spread, and inventing an uncertainty for it would dress a guess as a measurement.

7.3 Comparing drift models against a fixed environment

Tick several classes and press **COMPARE TICKED CLASSES**. Each is run from the same last known position, the same time of loss, the same wind and swell, and the same cached forecast — the **ONLY** thing that changes between rows is the drift model, so every difference in the table is attributable to it. Each alternative datum is drawn on the chart with its own circle, in outline, under the one in use.

This is the honest use of a taxonomy with no entry for this vessel

The class is an analogue. How far the answer moves when you pick a different one is therefore a result in its own right, and usually a larger one than any decimal place in the datum. On the supplied position the spread between classes runs to about half the search radius — the same order as the uncertainty within any one of them. The table says so in words underneath, so the comparison cannot be read as four competing answers when it is really one answer and its sensitivity.

The run is done in a single request rather than one per class, so the forecast is loaded once for all of them. That is not only speed: reloading it per class would invite exactly the environmental drift the comparison is trying to rule out.

A tidal current reverses about every six hours, so a hull adrift in one loops and creeps rather than running in a straight line. Over a day the displacement is set by the residual flow and the wind, not by the peak

current — which is why the radius can exceed the distance travelled, and why a drift prediction is far more sensitive to a small steady error than a passage ETA is.

Two exports sit under the result: CSV of the datum table, and GeoJSON carrying the last known position, the datum track, every hourly datum with its radius, and each ensemble endpoint. Both are built in the browser from what is on screen, so what gets handed over is exactly what was being looked at.

It refuses rather than guessing

If no cached forecast reaches the last known position, the prediction is refused outright with the reason — because the alternative is a wind-only track formatted to look exactly like a real one. Fetch a model that covers the position, and try again.

8 Exporting and saving

Every save of a calculated mission carries a UTC timestamp in its name —

Lewes_offshore_20260814T0317Z_shp.zip — so repeated saves of the same line do not overwrite each other and sort chronologically. The stamp reaches inside a zipped shapefile too, so two exports extracted into the same folder do not collide either.

"Save to" is ON by default, so pressing a format writes the file into docs/missions and the status bar reports the path. Untick it to get a browser download instead. Same bytes and the same name either way; only the destination differs.

The Export panel appears once you have calculated

It lives with the results. If you have drawn a line but not pressed CALCULATE, there is nothing to export yet.

Format	Use it for
Shapefile (.zip)	polyline + .prj/.dbf/.shx/.cpg
Shapefile — per leg (.zip)	one feature per leg, with results
GeoJSON	WGS84 lon/lat, track + waypoints
KML	Google Earth
GPX	route + track, for a chartplotter
CSV — waypoints	lat/lon, leg and cumulative NM
CSV — leg table	the computed transit

Shapefile CRS

GeoJSON, KML and GPX are defined as WGS84 longitude/latitude, so the CRS choice applies to the SHAPEFILE only. UTM writes projected metres with a matching .prj — usually what a survey package expects back. One zone is chosen for the whole line, because a shapefile carries exactly one .prj.

9 If something looks wrong

Symptom	Likely cause
Chart is blank	No tiles cached for this area and no network. Press Charts while connected. The graticule still gives you scale.
"current 0% covered"	No cycle cached, or none reaches this line. Press Currents.
A leg is flagged red	The cross-current exceeds what the vessel can crab out at that speed. The line cannot be flown as drawn at that speed — this is a refusal, not a slow leg.
Fuel looks implausibly low	Check the Weather setting is not on "ignore", and check the RPM window flag.
Port already in use	Another copy is running. Open it, or start this one with --port 8079.